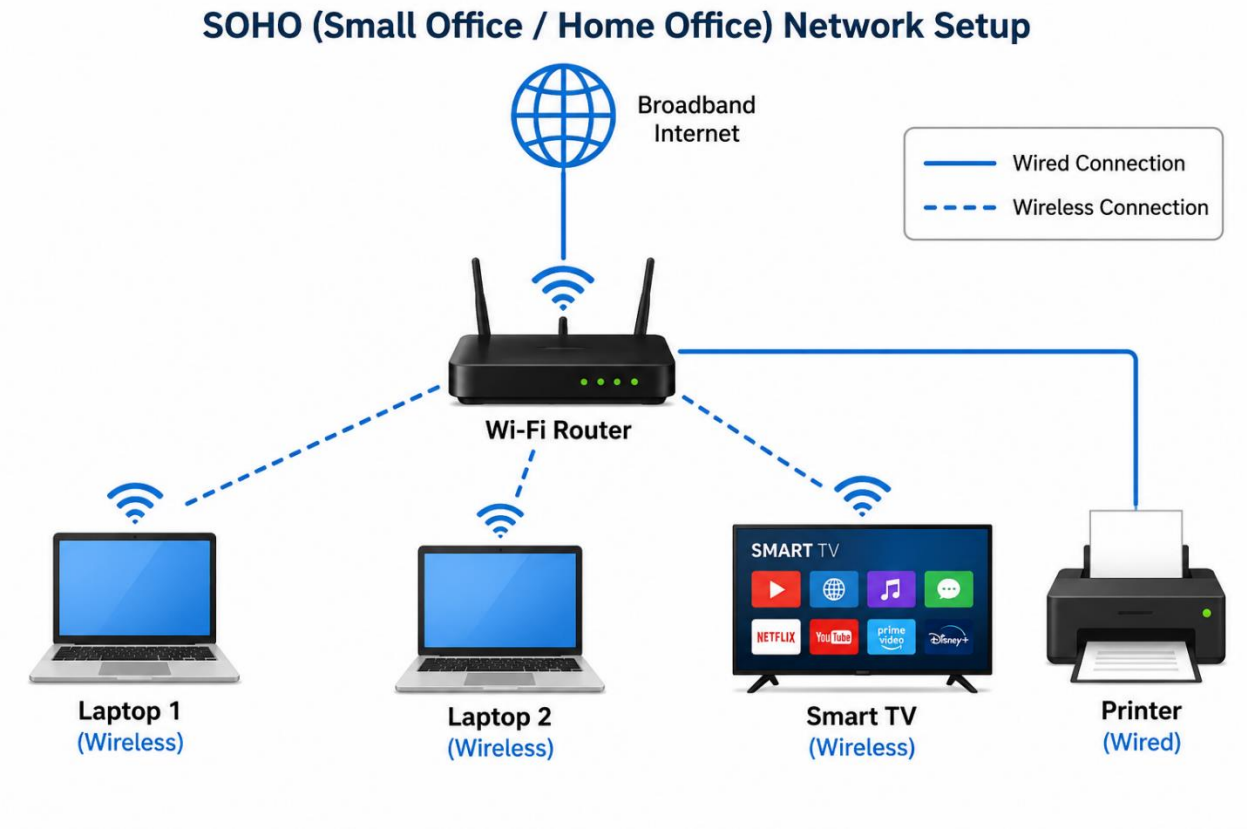
1.Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



2.List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Component	Suggested Brand/Model (India)
ISP Modem/ONT	ISP-provided Fiber ONT (Jio/Airtel/BSNL) or D-Link DSL-2750U (DSL)
Wi-Fi Router	TP-Link Archer C6 AC1200 Dual-Band Gigabit Router
Network Switch (if more wired ports are needed)	TP-Link LS1008 8-Port Desktop Switch
Wi-Fi Range Extender (optional)	TP-Link RE200 AC750 Range Extender
Ethernet Cables	D-Link Cat6 UTP Patch Cord (5 m / 10 m)
RJ45 Connectors (optional)	D-Link / MX Cat6 RJ45 Connectors
UPS (recommended)	APC Back-UPS 600VA / 1100VA

A simple physical connection is: ISP Fiber/DSL → Modem/ONT → TP-Link Archer C6 Router → TP-Link LS1008 Switch → Printer and Smart TV (wired), while Laptops connect via Wi-Fi. If Wi-Fi is weak in another room, place the TP-Link RE200 midway between the router and that area.

3.Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

- Connect the router

  Connect the ISP modem/ONT to the router's WAN/Internet port.

  Connect a laptop or PC to the router using Wi-Fi or a LAN cable.

- Open the router settings

  - Open a web browser.
  - Type 192.168.0.1 or 192.168.1.1 in the address bar.
  - Press Enter.

- Log in

  - Enter the default username and password (usually admin / admin).
  - Click Login.

- Change the default admin password

  - Go to Administration / Password settings.
  - Enter the old password.
  - Set a new strong password (example: Office@2026).
  - Click Save.

- Set a unique SSID (Wi-Fi name)

  - Go to Wireless Settings.
  - Enter a unique Wi-Fi name such as Maitri_SOHO_Network.
  - Click Save.

- Enable WPA2 security

  - Open Wireless Security.
  - Select WPA2-PSK (AES).
  - Enter a strong Wi-Fi password (example: Maitri@12345).
  - Click Save.

- Restart the router

  - Click Reboot/Restart.
  - Wait for the router to restart.

4.Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Device	MAC Address (Sample)	IP Assignment
Router (Gateway)	00:1A:2B:3C:4D:01	192.168.1.1
Printer (Static)	00:1A:2B:3C:4D:10	192.168.1.10
Smart TV (Static)	00:1A:2B:3C:4D:20	192.168.1.20
Laptop 1 (DHCP)	00:1A:2B:3C:4D:31	192.168.1.101
Laptop 2 (DHCP)	00:1A:2B:3C:4D:32	192.168.1.102

Notes

- Static IP devices: Printer and Smart TV always keep the same IP address so they can be easily found on the network.
- DHCP devices: Laptops automatically receive IP addresses from the router's DHCP server.
- Subnet Mask: 255.255.255.0 for all devices.
- Default Gateway: 192.168.1.1.
- DNS Server: 192.168.1.1 (or public DNS such as 8.8.8.8).

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5.Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

If one laptop cannot connect to the internet, I would check the following possible causes and solutions.

Possible Cause	Diagnosis and Fix
Wi-Fi is disconnected	Check if Wi-Fi is turned on and connected to the correct SSID. Reconnect the laptop to the Wi-Fi network.
Incorrect IP address	Run ipconfig and check the IP address. If needed, renew it using ipconfig /release and ipconfig /renew.
Router or modem problem	Check whether other devices have internet access. Restart the router and modem, then test the connection again.

Steps I Would Follow:

1. Check whether the laptop is connected to the correct Wi-Fi network.
2. Verify that the laptop has received a valid IP address from DHCP.
3. Restart the router and modem.
4. Test the internet connection by opening a website or using the `ping` command.